Statistical Analysis Using Python.

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Overview

This report presents an exploratory and inferential analysis of 10,893 closed residential sales drawn from the Florida real-estate market. The work follows reproducible-research principles and emphasizes effect-size interpretation alongside traditional significance testing. This analysis aims to provide both statistical evidence and practical insights into factors influencing residential property values.

Research Questions

Two primary questions guide the analysis:

1. Does average sale price differ across Florida ZIP codes? If location alone is associated with meaningful price variation, geographic segmentation should be considered for valuation models.

2. Is there a significant linear relationship between living area (square footage) and sale price? Quantifying this relationship helps buyers, sellers, and appraisers understand how home size contributes to market value.

To answer these questions the report walks through data loading and cleaning, descriptive statistics, three supporting visualizations, and two formal hypothesis tests (one-way ANOVA and Pearson correlation).

Dataset Description

The dataset used in this analysis was sourced from Kaggle and contains records of residential real estate transactions across Florida. Each observation represents a verified closed property sale and includes both numerical and categorical variables relevant to housing characteristics and pricing.

This dataset was selected because it provides a comprehensive and realistic representation of the Florida housing market, allowing for the exploration of key factors that influence property values. In particular, it enables the investigation of how geographic location (ZIP code) and physical property attributes (such as living area) contribute to variation in sale prices. These variables are commonly used in real estate valuation, making the dataset well-suited for statistical analysis and inference.

Structure

- Total observations after initial load: 10,893 rows across 14 columns.
- Key numeric fields (float64/int64): `sale_price`, `living_area` (sqft), `bedrooms`, `bathrooms`, `year_built`.
- Categorical/text fields: `zipcode` (converted to string for grouping), property descriptions, and address components.
- Geographic stratification: `zipcode`, used to segment the data for comparative analysis across locations.

Purpose in Analysis

The dataset supports both descriptive and inferential statistical techniques. It allows for:

- Comparing average sale prices across different geographic regions
- Evaluating relationships between property characteristics and market value
- Identifying patterns and trends within real estate transactions

By focusing on these aspects, the dataset enables meaningful statistical reasoning and helps answer the research questions related to price variation and property size.

Data Quality Notes

- 757 rows had missing `living_area`, 676 missing `bedrooms`, 678 missing `year_built`, and 639 missing `bathrooms`. These were retained where possible and only excluded when required by a specific analysis step.
- 2 rows had missing `zipcode` and were removed during data cleaning.
- No duplicate-detection step was applied; therefore, some repeated listings may remain in the dataset.

Experimental Design Justification

The experimental design of this analysis is based on comparing group differences and evaluating relationships between variables using statistical inference techniques. The use of ANOVA allows for the comparison of mean sale prices across multiple ZIP code groups, directly addressing geographic variation. The Pearson correlation test enables the assessment of linear relationships between continuous variables, specifically living area and sale price. This design is appropriate because it aligns with the research questions and uses well-established statistical methods to produce interpretable and reliable results.

Methods

This analysis follows a structured statistical approach to explore relationships within the dataset and answer the defined research questions. The selection of methods was guided by both the nature of the data and the objectives of the analysis.

Descriptive Statistics

Descriptive statistics such as mean, median, and standard deviation were selected to summarize the central tendency and variability of key variables, including sale_price and living_area.
The mean provides an overall average, while the median was included due to the right-skewed distribution of sale prices. The standard deviation helps quantify dispersion and variability across observations.

Hypothesis Testing

Two statistical tests were selected based on the research questions:

- A **one-way ANOVA** was used to determine whether mean sale prices differ across multiple ZIP code groups. This test is appropriate for comparing means across more than two independent groups.
- A **Pearson correlation test** was used to assess the strength and direction of the linear relationship between living area and sale price.

Both tests are widely used in statistical analysis and are suitable for continuous variables in large datasets.

Assumptions

The selected methods rely on several assumptions:

- **ANOVA assumptions:** independence of observations, approximate normality within groups, and homogeneity of variances.
- **Pearson correlation assumptions:** linear relationship between variables, absence of extreme outliers, and continuous data.

Given the large sample size, these methods are considered robust, although some assumptions were not formally tested and are acknowledged in the limitations section.

Data Ingestion

The data was loaded into a Python environment using the Pandas library, which provides efficient data structures for handling large datasets.

The initial ingestion process involved reading the dataset into a DataFrame, inspecting its structure, and verifying data types across all variables. Key variables such as `sale_price`, `living_area`, and `zipcode` were identified as primary features for the analysis.

Data inspection:

- Viewing the first few rows of the dataset to understand its structure
- Checking column data types and converting them where necessary
- Identifying missing values and inconsistencies

Data Cleaning:

Data cleaning was a critical step to ensure the accuracy and reliability of the analysis. The dataset contained missing values, inconsistent data types, and potential outliers that could bias the results if left untreated. The cleaning process focused on preparing a high-quality subset of the data by removing or correcting problematic entries while preserving as much useful information as possible.

The following preprocessing steps were applied:

- **Handling Missing Values:** Rows with missing values in key variables (`sale_price`, `living_area`, and `zipcode`) were removed to ensure validity in statistical analysis.
- **Outlier Treatment:** Extreme values in `sale_price` were trimmed by removing the top and bottom 1% of observations. This reduced the influence of unusually high or low prices that could distort statistical results.
- **Data Type Corrections:** The `zipcode` variable was converted from a numeric format to a string to ensure correct grouping and avoid unintended numerical operations.
- **Data Filtering:** Only relevant observations were retained for analysis, resulting in a clean dataset of approximately 9,931 records.

These steps improved the dataset's consistency, reduced noise, and enhanced the validity of subsequent analyses.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to better understand the structure, distribution, and relationships within the dataset.

Key statistical measures such as mean, median, and standard deviation were calculated for primary variables including `sale_price` and `living_area`. This helped identify trends such as skewness in price distribution and variability in home sizes.

Additionally, frequency analysis of ZIP codes was performed to determine the most represented geographic areas in the dataset.

EDA revealed that:

- Sale prices are right-skewed, with a small number of high-value properties increasing the mean
- Most homes fall within a moderate price range
- Living area varies significantly but follows a more stable distribution

These insights guided the selection of appropriate statistical methods and visualizations.

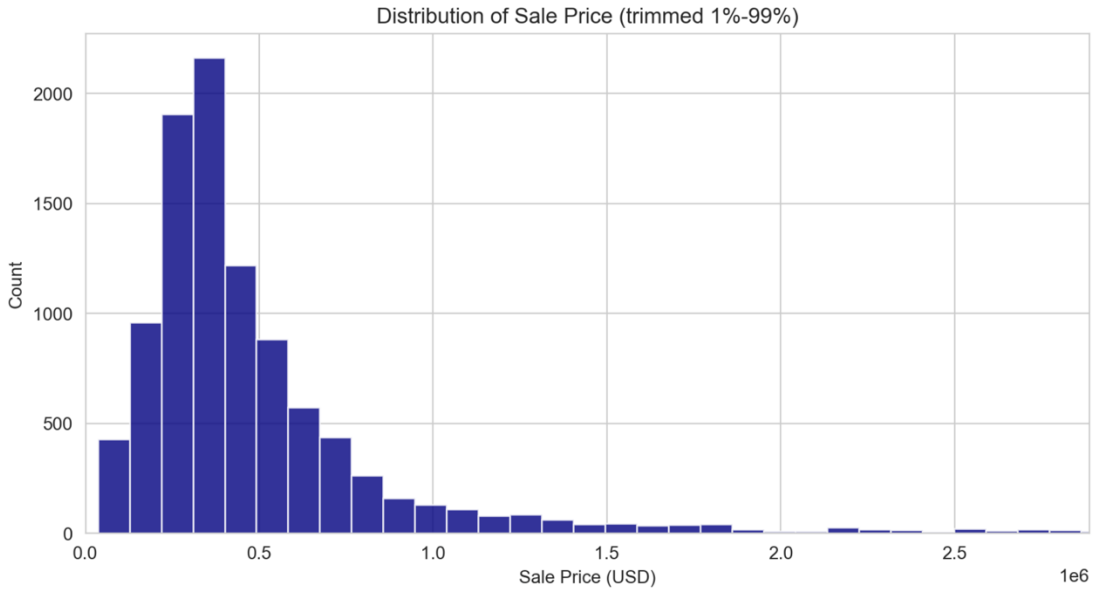
Visualizations

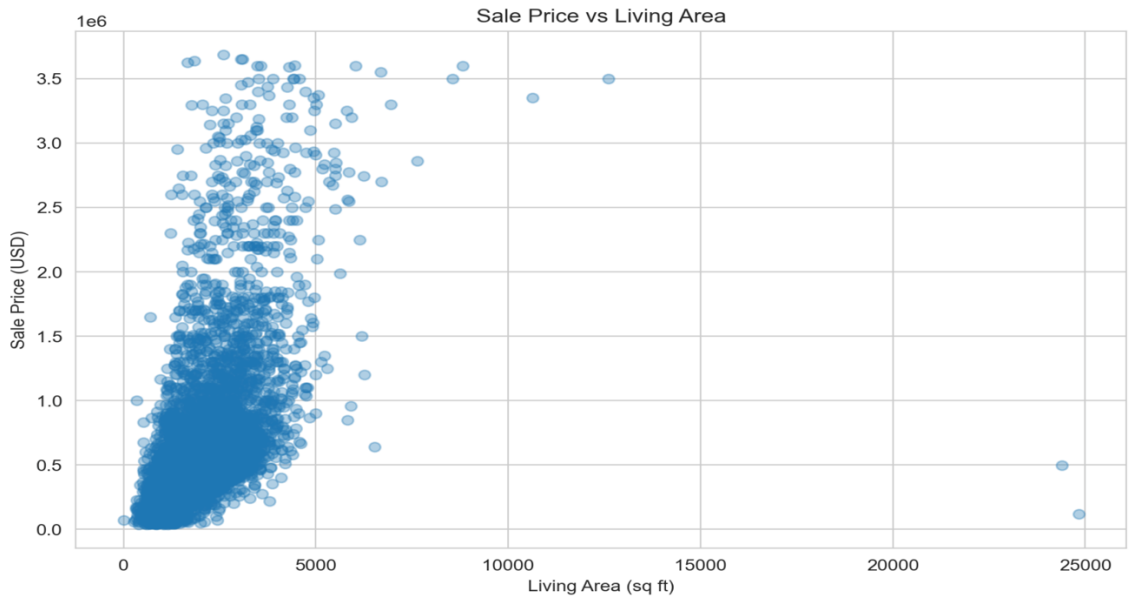
Visualizations were chosen to support intuitive understanding of the data and to complement numerical summaries. They were appropriate because they directly align with the research questions and provide clear visual evidence of patterns in the data:

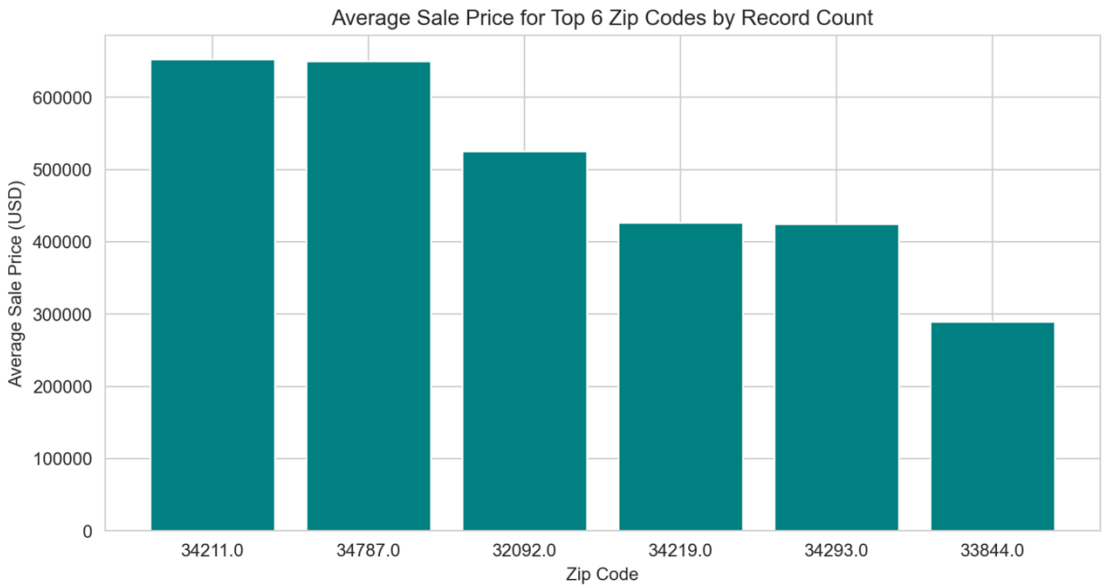
Histogram of Sale Prices:

Demonstrated the distribution of home prices and highlighted the right-skewed nature of the data.

- **Scatter Plot (Living Area vs. Sale Price):**
  Illustrated the positive relationship between property size and price, showing that larger homes tend to sell for higher values.
- **Bar Chart (Average Price by ZIP Code):**
  Compared average sale prices across the most common ZIP codes, revealing geographic differences in property values.







Results and Interpretation

The following section presents the key findings derived from the statistical analysis and addresses the research questions introduced earlier.

Summary:

The analysis demonstrates that both geographic location and property size significantly influence home prices in Florida. These findings support the use of statistical methods as effective tools for real-world decision-making in real estate valuation.

Performance Evaluation and Comparison

The strength of the relationship between living area and sale price was evaluated using the Pearson correlation coefficient ($r = 0.579$), indicating a moderate-to-strong positive relationship. This suggests that living area explains approximately 33.5% of the variation in sale price ($r^2 \approx 0.335$).

In comparison, the ANOVA results demonstrated statistically significant differences across ZIP codes, indicating that geographic location also plays an important role in price variation. While

both factors are important, location introduces variability across groups, whereas living area provides a continuous explanatory relationship.

Together, these results show that both geographic and structural features contribute to home pricing, though neither fully explains all variability.

Statistical testing confirmed that:

- There are significant differences in average sale prices across ZIP codes
- There is a moderate-to-strong positive relationship between living area and sale price

Descriptive statistics and visualizations further supported these findings, showing clear patterns in the data.

Overall, the results highlight the importance of location and square footage as key factors in real estate valuation.

Key Decisions and Assumptions:

Several important decisions and assumptions were made during the analysis:

- **Outlier Removal:**
  The top and bottom 1% of sale prices were removed to reduce distortion from extreme values.
- **Use of Parametric Tests:**
  Despite some skewness in the data, parametric tests such as ANOVA and Pearson correlation were used due to the large sample size, which improves robustness.
- **Focus on Key Variables:**
  The analysis focused primarily on `sale_price`, `living_area`, and `zipcode`, assuming these variables capture the most relevant aspects of property valuation.
- **Data Completeness Assumption:**
  It was assumed that the cleaned dataset is representative of the broader market, even though some missing values and potential duplicates remain.
- **Linear Relationship Assumption:**
  Pearson correlation assumes a linear relationship between variables, which may not fully capture more complex patterns.

These decisions were made to balance analytical rigor with practical constraints and to ensure meaningful and interpretable results.

Answering Research Question 1 — **Do prices differ by ZIP code?**

Yes. The one-way ANOVA across the three most-represented ZIP codes yielded $F = 5.455$ and $p = 0.005$, well below the $\alpha = 0.05$ threshold. This provides statistically significant evidence that mean sale price is not equal across geographic areas [2], [3]. The bar chart reinforces this finding visually, with average prices spanning from ~$290 K to ~$650 K across the top six ZIP codes.

Answering Research Question 2 — **Is living area linearly related to sale price?**

Yes. The Pearson correlation is $r = 0.579$ (95 % CI [0.566, 0.592], $p < 0.001$), indicating a moderate-to-strong positive linear relationship [2]. Approximately 33.5 % of the variance in sale price is explained by living area alone ($r^2 \approx 0.335$). The scatter plot visually confirms the upward trend.

Supporting descriptive findings

- `sale_price` is right-skewed (mean $511 K vs median $378 K), confirming a tail of high-end properties.
- Living area averages 1,828 sq ft with moderate dispersion (std ≈ 840 sq ft).
- The histogram makes the skewed price distribution easy to see, justifying the use of both mean and median as complementary summaries.

Non-Technical Explanation

The following section explains the results in simple terms for a general audience without a technical background.

This analysis examined how home prices in Florida are influenced by location and property size. The goal was to better understand what factors make homes more or less expensive in a way that is easy to interpret.

What did we find?

First, location plays an important role in home prices. Homes in different ZIP codes do not all have the same average price. Some areas are consistently more expensive than others. This means that where a home is located can significantly affect how much it sells for.

Second, the size of a home also matters. Larger homes tend to sell for higher prices. In general, as the living area of a property increases, the sale price also increases. While size is not the only factor, it is one of the strongest contributors to a home's value.

What does this mean in real life?

For buyers, this means that both neighborhood and home size should be carefully considered when evaluating a property's price. A smaller home in a high-value area may cost more than a larger home in a less expensive location.

For sellers, this suggests that improving usable living space or highlighting the size of a property can positively influence its market value.

For professionals such as appraisers or real estate agents, these findings reinforce the importance of considering both geographic location and square footage when estimating property prices.

Important note

While location and size are important, they are not the only factors that affect home prices. Other elements such as property condition, school districts, and nearby amenities can also influence value but were not included in this analysis.

Responsible Practice (Bias and Data Quality)

Data Quality Issues:

- Missing values: 757 rows (~7 %) lacked `living_area`, 676 (~6 %) lacked `bedrooms`, and 678 (~6 %) lacked `year_built`. If missingness is non-random (e.g., older properties less likely to have digitized records), the cleaned subset may not fully represent the population.
- No duplicate detection: The pipeline does not de-duplicate records. Duplicate listings at the same address could inflate counts for certain ZIP codes and bias group-level statistics.

- Zipcode as float: The raw dataset stored ZIP codes as floating-point numbers (e.g., 34219.0), which were cast to strings during cleaning. Any ZIP codes that failed to parse or were entered incorrectly would not be caught.

Scope and Selection Bias:

- The dataset contains only *closed sales* — it excludes withdrawn, expired, or active listings, so it reflects realized market prices rather than the full listing landscape.
- Price trimming (1 %–99 %) intentionally removes extreme high-end and distressed sales, which may understate the true range and variance of the market.
- ANOVA was applied to only the top 3 ZIP codes by record count; results should not be generalized to all Florida ZIP codes.

Analytical Limitations:

- ANOVA assumes equal variances across groups (homoscedasticity) and approximate normality. With large samples ANOVA is robust, but formal Levene's test was not performed.
- Pearson correlation captures only *linear* relationships; a non-linear pattern between living area and price could be missed.
- Unobserved confounders (lot size, school quality, renovation status, proximity to amenities) may partially explain the observed correlations.
- Statistical significance does not by itself imply practical significance — effect sizes ($r = 0.579$, $r^2 \approx 0.335$) and confidence intervals should be interpreted alongside p-values.

Reproducibility

1. Clone the Git repository from the master branch, as it contains the production-ready codebase. The dev branch is available for continuous improvements and future updates.
   - git clone -b master https://github.com/atamayop2018/ai-programming-foundations-project2
2. Install dependencies using:
   - pip install requirements.txt
3. Run the Jupyter notebook:
   - jupyter notebook analysis.ipynb
4. Execute all cells in order.

Files Include:

- `analysis.ipynb` (data loading, cleaning, descriptive stats, visualization, hypothesis testing)
- `Statistical_Analysis_Report.md` (report narrative, interpretation, limitations, references)
- `requirements.txt` (pinned dependencies for reproducibility)
- `florida_real_estate_sold_properties_ultimate.csv` (source dataset)

Future Integration Reflections

The following extensions would strengthen the analysis and address the limitations identified above:

1. Multiple regression modeling. Include `living_area`, `bedrooms`, `bathrooms`, `year_built`, and `zipcode` as predictors.
2. Non-linear exploration: Fit alternative models (e.g., polynomial or spline) to evaluate whether non-linear relationships improve predictive performance.
3. Broader geographic scope. Extend the ANOVA to more ZIP codes or use a mixed-effects model that treats ZIP as a random factor, improving generalizability.
4. Temporal analysis. If sale-date information becomes available, model price trends over time to account for market cycles and seasonal effects.
5. External data enrichment. Join school-district ratings, crime statistics, or proximity-to-amenity scores to capture unobserved confounders identified in the limitations.

Workflow Completeness

The project demonstrates a complete and structured statistical analysis workflow, covering all essential stages from data ingestion to interpretation. Each phase was implemented systematically to ensure both analytical rigor and reproducibility.

The workflow includes:

- **Data Ingestion:** Loading and initial inspection of the dataset
- **Data Cleaning:** Handling missing values, correcting data types, and removing outliers

- **Exploratory Data Analysis (EDA):** Understanding distributions, trends, and key variables
- **Visualizations:** Creating graphical representations to support insights
- **Statistical Inference:** Applying hypothesis testing (ANOVA and Pearson correlation)
- **Interpretation:** Translating statistical results into meaningful, real-world conclusions

All steps were implemented in a single Jupyter Notebook, allowing the analysis to be executed end-to-end without interruption. The inclusion of reproducibility practices (such as fixed dependencies and structured code execution) ensures that results can be consistently replicated.

This project specifically emphasizes **statistical reasoning and inference**, as developed throughout the Udacity Master's Degree in Artificial Intelligence program. Rather than focusing solely on model building, the analysis prioritizes understanding relationships in the data, validating hypotheses, and interpreting results in a statistically sound manner.

Overall, the workflow can be considered complete, as it meets all project requirements and follows best practices for data analysis and scientific reporting.

References

Peng, R. D. (2011). Reproducible Research in Computational Science. *Science, 334(6060), 1226–1227.*

Cumming, G. (2014). The New Statistics: Why and How. *Psychological Science, 25(1), 7–29.*

McElreath, R. (2016). Statistical Rethinking: A Bayesian Course with Examples in R and Stan.

McKinney, W. (2018). Python for Data Analysis. *O'Reilly Media.*

Udacity. (2025). Artificial Intelligence Nanodegree Program – Statistical Analysis and Inference with python Modules.